

FMUSER FUQ-4647R HD Encoder Modulator 8-24 HDMI 1 ASI 1 USB 128 IP Input MPEG-4 AVC-H264 Encoding Multiplexing DVB-C-T-ATSC-ISDB-T Output CATV Headend Hotel TV System Modulator



Product Overview

The FMUSER FUQ-4647B is an encoder modulator integrating encoding, multiplexing, and modulation in a 1RU chassis. It processes inputs including 8/16/24 HDMI, 1 ASI, 1 USB player, and 128 IP streams, encoding with MPEG-4 AVC/H.264 and outputting 12 DVB-C RF carriers plus ASI/IP mirrors. Featuring web-based control for remote management, it is ideal for compact CATV headends and hospitality TV systems.

Key feature

Supports multi-format inputs: 8/16/24 HDMI, 1 ASI, 1 USB player, and 128 IP streams via GE port.

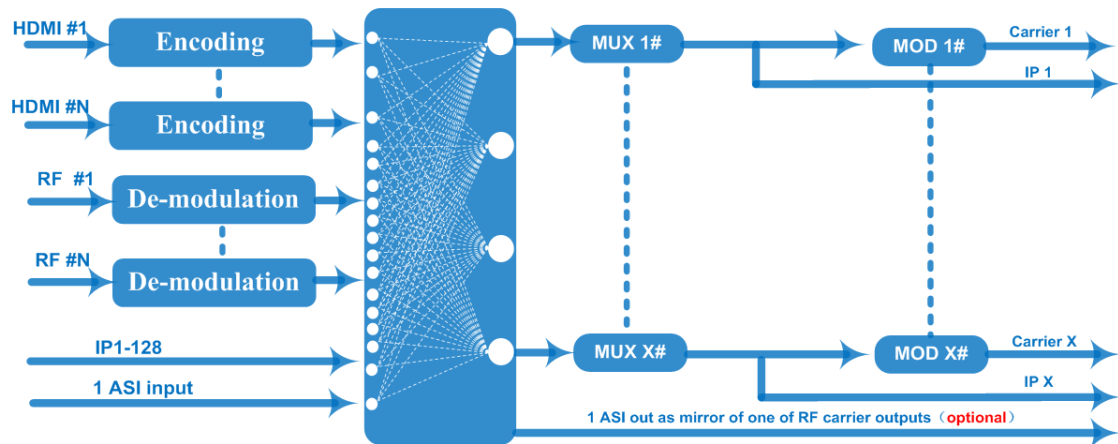
Offers flexible modulation: Outputs 12 DVB-C RF carriers (or optional formats) with $\leq 192\text{MHz}$ bandwidth.

Provides advanced processing: MPEG-4 AVC/H.264 encoding, PID remapping, PCR adjustment, and PSI/SI editing.

Includes dual monitoring: ASI output mirror and MPTS IP streaming via UDP/RTP/RTSP.

Features web-based control for remote management and firmware updates.

Principle Chart



Technical Specifications

Input	8/16/24 HDMI inputs for option (or HDMI and different Tuner mix input, details pls check below order guide) 1 ASI in for re-mux 1 USB Player input for re-mux ("xxx.ts" video) 128 IP input over UDP and RTP, GE port, RJ45	8/16/24 HDMI inputs for option (or HDMI and different Tuner mix input, details pls check below order guide) 1 ASI in for re-mux 1 USB Player input for re-mux ("xxx.ts" video) 128 IP input over UDP and RTP, GE port, RJ45	8/16/24 HDMI inputs for option (or HDMI and different Tuner mix input, details pls check below order guide) 1 ASI in for re-mux 1 USB Player input for re-mux ("xxx.ts" video) 128 IP input over UDP and RTP, GE port, RJ45	8/16/24 HDMI inputs for option (or HDMI and different Tuner mix input, details pls check below order guide) 1 ASI in for re-mux 1 USB Player input for re-mux ("xxx.ts" video) 128 IP input over UDP and RTP, GE port, RJ45	8/16/24 HDMI inputs for option (or HDMI and different Tuner mix input, details pls check below order guide) 1 ASI in for re-mux 1 USB Player input for re-mux ("xxx.ts" video) 128 IP input over UDP and RTP, GE port, RJ45	8/16/24 HDMI inputs for option (or HDMI and different Tuner mix input, details pls check below order guide) 1 ASI in for re-mux 1 USB Player input for re-mux ("xxx.ts" video) 128 IP input over UDP and RTP, GE port, RJ45
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Video (HDMI Input)	Resolution	Input	Input	Input	1920×1080_60P, 1920×1080_60i, 1920×1080_50P, 1920×1080_50i, 1280×720_60P, 1280×720_50P, 720×576_50i, 720×480_60i,	1920×1080_60P, 1920×1080_60i, 1920×1080_50P, 1920×1080_50i, 1280×720_60P, 1280×720_50P, 720×576_50i, 720×480_60i,
Video (HDMI Input)	Resolution	Output	Output	Output	1920×1080_30P, 1920×1080_25P, 1280×720_30P, 1280×720_25P, 720×576_25P, 720×480_30P,	1920×1080_30P, 1920×1080_25P, 1280×720_30P, 1280×720_25P, 720×576_25P, 720×480_30P,
Video (HDMI Input)	Encoding	MPEG-4 AVC/H.264	MPEG-4 AVC/H.264	MPEG-4 AVC/H.264	MPEG-4 AVC/H.264	MPEG-4 AVC/H.264
Video (HDMI Input)	Bit-rate	1Mbps~13 Mbps each channel	1Mbps~13 Mbps each channel	1Mbps~13 Mbps each channel	1Mbps~13 Mbps each channel	1Mbps~13 Mbps each channel
Video (HDMI Input)	Rate Control	CBR/VBR	CBR/VBR	CBR/VBR	CBR/VBR	CBR/VBR
Video (HDMI Input)	GOP Structure	IP...P (P Frame adjustment, without B Frame)	IP...P (P Frame adjustment, without B Frame)	IP...P (P Frame adjustment, without B Frame)	IP...P (P Frame adjustment, without B Frame)	IP...P (P Frame adjustment, without B Frame)
Audio (HDMI Input)	Encoding	MPEG-1 Layer 2, LC-AAC, HE-AAC and AC3 Pass through	MPEG-1 Layer 2, LC-AAC, HE-AAC and AC3 Pass through	MPEG-1 Layer 2, LC-AAC, HE-AAC and AC3 Pass through	MPEG-1 Layer 2, LC-AAC, HE-AAC and AC3 Pass through	MPEG-1 Layer 2, LC-AAC, HE-AAC and AC3 Pass through

Audio (HDMI Input)	Sampling rate	48KHz	48KHz	48KHz	48KHz	48KHz
Audio (HDMI Input)	Resolution	24-bit	24-bit	24-bit	24-bit	24-bit
Audio (HDMI Input)	Audio Gain	0-255 Adjustable	0-255 Adjustable	0-255 Adjustable	0-255 Adjustable	0-255 Adjustable
Audio (HDMI Input)	MPEG-1 Layer 2 Bit-rate	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps
Audio (HDMI Input)	LC-AAC Bit-rate	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps	48/56/64/80/96/112/128/160/192/224/256/320/384 kbps
Audio (HDMI Input)	HE-AAC Bit-rate	48/56/64/80/96/112/128 kbps	48/56/64/80/96/112/128 kbps	48/56/64/80/96/112/128 kbps	48/56/64/80/96/112/128 kbps	48/56/64/80/96/112/128 kbps
Tuner Section	DVB-S	Input Frequency	Input Frequency	950-2150M Hz	950-2150M Hz	950-2150M Hz
Tuner Section	DVB-S	Symbol Rate	Symbol Rate	0.5~45 Msps	0.5~45 Msps	0.5~45 Msps
Tuner Section	DVB-S	Constellation	Constellation	QPSK	QPSK	QPSK
Tuner Section	DVB-S	FEC	FEC	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Tuner Section	DVB-S2	Input Frequency	Input Frequency	950-2150M Hz	950-2150M Hz	950-2150M Hz
Tuner Section	DVB-S2	Symbol Rate	Symbol Rate	QPSK/8PSK/16APSK: 0.5~45 Msps 32APSK: 0.5~40 Msps	QPSK/8PSK/16APSK: 0.5~45 Msps 32APSK: 0.5~40 Msps	QPSK/8PSK/16APSK: 0.5~45 Msps 32APSK: 0.5~40 Msps

Tuner Section	DVB-S2	Constellation	Constellation	QPSK/8PSK/16APSK/32APSK	QPSK/8PSK/16APSK/32APSK	QPSK/8PSK/16APSK/32APSK
Tuner Section	DVB-S2	FEC	FEC	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 32APSK: 3/4, 4/5, 5/6, 8/9	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 32APSK: 3/4, 4/5, 5/6, 8/9	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 32APSK: 3/4, 4/5, 5/6, 8/9
Tuner Section	DVB-S2	Max input bitrate per tuner	Max input bitrate per tuner	≤170 Mbps	≤170 Mbps	≤170 Mbps
Tuner Section	DVB-S2X	Symbol Rate	Symbol Rate	QPSK/8PSK/16APSK: 0.5~45 Msps; 8APSK/32APSK: 0.5~40 Msps	QPSK/8PSK/16APSK: 0.5~45 Msps; 8APSK/32APSK: 0.5~40 Msps	QPSK/8PSK/16APSK: 0.5~45 Msps; 8APSK/32APSK: 0.5~40 Msps
Tuner Section	DVB-S2X	Constellation	Constellation	QPSK/8PSK/16APSK/8APSK/32APSK	QPSK/8PSK/16APSK/8APSK/32APSK	QPSK/8PSK/16APSK/8APSK/32APSK

Tuner Section	DVB-S2X	FEC	FEC	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10, 23/36, 25/36, 13/18 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10, 23/36, 25/36, 13/18 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10, 23/36, 25/36, 13/18 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9
Tuner Section	DVB-S2X	Max input bitrate per tuner	Max input bitrate per tuner	≤170 Mbps	≤170 Mbps	≤170 Mbps
Tuner Section	Signal Strength	-65 ~ -25dBm	-65 ~ -25dBm	-65 ~ -25dBm	-65 ~ -25dBm	-65 ~ -25dBm
Multiplexing	Maximum PID Remapping	255 input per channel	255 input per channel	255 input per channel	255 input per channel	255 input per channel

Multiplexing	Function	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)
Multiplexing	Function	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting
Multiplexing	Function	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically
Modulation	DVB-C	QAM Channel	QAM Channel	QAM Channel	12 non-adjacent carriers output (maximum bandwidth 192MHz)	12 non-adjacent carriers output (maximum bandwidth 192MHz)
Modulation	DVB-C	Standard	Standard	Standard	EN300 429/ITU-T J.83A/B	EN300 429/ITU-T J.83A/B
Modulation	DVB-C	MER	MER	MER	≥40db	≥40db
Modulation	DVB-C	RF frequency	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	DVB-C	RF output level	RF output level	RF output level	-20~+3dbm, 0.1db step	-20~+3dbm, 0.1db step
Modulation	DVB-C	Symbol Rate	Symbol Rate	Symbol Rate	5.0Msps~7.0Msps, 1ksps stepping	5.0Msps~7.0Msps, 1ksps stepping
Modulation	DVB-C				J.83A	J.83B
Modulation	DVB-C	Constellation	Constellation	Constellation	16/32/64/128/256QAM	64/256 QAM
Modulation	DVB-C	Bandwidth	Bandwidth	Bandwidth	8M	6M
Modulation	DVB-T	Standard	Standard	Standard	EN300744	EN300744
Modulation	DVB-T	FFT mode	FFT mode	FFT mode	2K,4K, 8K	2K,4K, 8K

Modulation	DVB-T	Bandwidth	Bandwidth	Bandwidth	6M, 7M, 8M	6M, 7M, 8M
Modulation	DVB-T	Constellation	Constellation	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulation	DVB-T	Guard Interval	Guard Interval	Guard Interval	1/4, 1/8, 1/16, 1/32	1/4, 1/8, 1/16, 1/32
Modulation	DVB-T	FEC	FEC	FEC	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Modulation	DVB-T	MER	MER	MER	≥42 dB	≥42 dB
Modulation	DVB-T	RF frequency	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	DVB-T	RF out	RF out	RF out	8 (or 16) non-adjacent carriers output (maximum bandwidth 192MHz)	8 (or 16) non-adjacent carriers output (maximum bandwidth 192MHz)
Modulation	DVB-T	RF output level	RF output level	RF output level	-20~+3dBm, 0.1db step	-20~+3dBm, 0.1db step
Modulation	ATSC	Standard	Standard	Standard	ATSC A/53	ATSC A/53
Modulation	ATSC	Bandwidth	Bandwidth	Bandwidth	6M	6M
Modulation	ATSC	Constellation	Constellation	Constellation	8VSB	8VSB
Modulation	ATSC	FEC	FEC	FEC	RS(208 18 8)+Trellis	RS(208 18 8)+Trellis
Modulation	ATSC	MER	MER	MER	≥40dB	≥40dB
Modulation	ATSC	ACL	ACL	ACL	-55 dBc	-55 dBc
Modulation	ATSC	RF frequency	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step

Modulation	ATSC	RF out	RF out	RF out	8 non-adjacent carriers output (maximum bandwidth 192MHz)	8 non-adjacent carriers output (maximum bandwidth 192MHz)
Modulation	ATSC	RF output level	RF output level	RF output level	-20~+3dbm (for all carriers), 0.5db stepping	-20~+3dbm (for all carriers), 0.5db stepping
Modulation	ISDB-T	Standard	Standard	Standard	ARIB STD-B31	ARIB STD-B31
Modulation	ISDB-T	Bandwidth	Bandwidth	Bandwidth	6M	6M
Modulation	ISDB-T	Constellation	Constellation	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulation	ISDB-T	Guard Interval	Guard Interval	Guard Interval	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4
Modulation	ISDB-T	Transmission Mode	Transmission Mode	Transmission Mode	2K, 4K, 8K	2K, 4K, 8K
Modulation	ISDB-T	Code rate	Code rate	Code rate	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Modulation	ISDB-T	MER	MER	MER	≥40dB	≥40dB
Modulation	ISDB-T	RF frequency	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	ISDB-T	RF out	RF out	RF out	6 (or 16) non-adjacent carriers output (maximum bandwidth 192MHz)	6 (or 16) non-adjacent carriers output (maximum bandwidth 192MHz)
Modulation	ISDB-T	RF output level	RF output level	RF output level	-20dBm~+3dBm, 0.1dB stepping	-20dBm~+3dBm, 0.1dB stepping

Stream output	1 ASI output as mirror of one of RF output carriers (Optional) 12 MPTS output over UDP and RTP/RTSP as mirror of 12 DVB-C carriers, 8/16 MPTS output over UDP and RTP/RTSP as mirror of 8/16 DVB-T carriers (Optional) 8 MPTS output over UDP and RTP/RTSP as mirror of 8 ATSC carriers (Optional) 6/16 MPTS output over UDP and RTP/RTSP as mirror of 6/16 ISDB-T carriers (Optional) 1*1000M Base-T Ethernet interface, GE port	1 ASI output as mirror of one of RF output carriers (Optional) 12 MPTS output over UDP and RTP/RTSP as mirror of 12 DVB-C carriers, 8/16 MPTS output over UDP and RTP/RTSP as mirror of 8/16 DVB-T carriers (Optional) 8 MPTS output over UDP and RTP/RTSP as mirror of 8 ATSC carriers (Optional) 6/16 MPTS output over UDP and RTP/RTSP as mirror of 6/16 ISDB-T carriers (Optional) 1*1000M Base-T Ethernet interface, GE port	1 ASI output as mirror of one of RF output carriers (Optional) 12 MPTS output over UDP and RTP/RTSP as mirror of 12 DVB-C carriers, 8/16 MPTS output over UDP and RTP/RTSP as mirror of 8/16 DVB-T carriers (Optional) 8 MPTS output over UDP and RTP/RTSP as mirror of 8 ATSC carriers (Optional) 6/16 MPTS output over UDP and RTP/RTSP as mirror of 6/16 ISDB-T carriers (Optional) 1*1000M Base-T Ethernet interface, GE port	1 ASI output as mirror of one of RF output carriers (Optional) 12 MPTS output over UDP and RTP/RTSP as mirror of 12 DVB-C carriers, 8/16 MPTS output over UDP and RTP/RTSP as mirror of 8/16 DVB-T carriers (Optional) 8 MPTS output over UDP and RTP/RTSP as mirror of 8 ATSC carriers (Optional) 6/16 MPTS output over UDP and RTP/RTSP as mirror of 6/16 ISDB-T carriers (Optional) 1*1000M Base-T Ethernet interface, GE port	1 ASI output as mirror of one of RF output carriers (Optional) 12 MPTS output over UDP and RTP/RTSP as mirror of 12 DVB-C carriers, 8/16 MPTS output over UDP and RTP/RTSP as mirror of 8/16 DVB-T carriers (Optional) 8 MPTS output over UDP and RTP/RTSP as mirror of 8 ATSC carriers (Optional) 6/16 MPTS output over UDP and RTP/RTSP as mirror of 6/16 ISDB-T carriers (Optional) 1*1000M Base-T Ethernet interface, GE port	1 ASI output as mirror of one of RF output carriers (Optional) 12 MPTS output over UDP and RTP/RTSP as mirror of 12 DVB-C carriers, 8/16 MPTS output over UDP and RTP/RTSP as mirror of 8/16 DVB-T carriers (Optional) 8 MPTS output over UDP and RTP/RTSP as mirror of 8 ATSC carriers (Optional) 6/16 MPTS output over UDP and RTP/RTSP as mirror of 6/16 ISDB-T carriers (Optional) 1*1000M Base-T Ethernet interface, GE port
System function	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)

System function	Chinese and English language	Chinese and English language	Chinese and English language	Chinese and English language	Chinese and English language	Chinese and English language
System function	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade
Miscellaneous	Dimension (W×L×H)	Dimension (W×L×H)	482mm×328mm×44mm	482mm×328mm×44mm	482mm×328mm×44mm	482mm×328mm×44mm
Miscellaneous	Environment	Environment	0~45℃(work); -20~80℃ (Storage)	0~45℃(work); -20~80℃ (Storage)	0~45℃(work); -20~80℃ (Storage)	0~45℃(work); -20~80℃ (Storage)
Miscellaneous	Power requirements	Power requirements	AC 110V±10%, 50/60Hz, AC 220 ±10%, 50/60Hz	AC 110V±10%, 50/60Hz, AC 220 ±10%, 50/60Hz	AC 110V±10%, 50/60Hz, AC 220 ±10%, 50/60Hz	AC 110V±10%, 50/60Hz, AC 220 ±10%, 50/60Hz

Ordering Information

Input Section:

	HDMI	DVB-S2/S2X	DVB-C/T/T2/ISDB-T
Option 1	8	/	/
Option 2	16	/	/
Option 3	24	/	/
Option 4	8	8	/
Option 5	8	16	/
Option 6	16	8	/
Option 7	8	/	8
Option 8	8	/	16
Option 9	16	/	8
Option 10	8	8	8
Option 11	16	4	4

Output Section:

RF output standard	RF output standard	IP output	RF carrier channels	Scrambling
DVB-C out	DVB-C out	12	12	N/A
DVB-T out	V1	8	8	N/A
DVB-T out	V2	16	16	N/A
ATSC out	ATSC out	8	8	N/A
ISDB-T out	V1	6	6	N/A
ISDB-T out	V2	16	16	N/A

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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